Data Networks

HW 1: Physical Layer

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Question 1

Compare the delay in sending an x-bit message over a k-hop path in a circuit-switched network and in a (lightly loaded) packet-switched network. The circuit setup time is s sec, the propagation delay is d sec per hop, the packet size is p bits, and the data rate is b bps. Under what conditions does the packet network have a lower delay? Also, explain the conditions under which a packet-switched network is preferable to a circuit switched network.

Solution

With circuit switching, at $t = s$ the circuit is set up, at $t = s + x/b$ the last bit is sent, at $t = s + x/b + kd$ the message arrives.

$t_c = s + x/b + kd$

In packet switching, the last bit is sent at $t = x/b$. To get to the final destination, the last packet must be retransmitted k-1 times by intermediate routers, with each retransmission taking p/b sec, so the total delay is $x/b + (k-1)p/b + kd$.

$t_p = x/b + (k-1)p/b + kd$

Packet switching is faster if $s > (k-1)p/b$. In addition to the faster transmission under these conditions, packet switching is preferable when fault-tolerant transmission in the presence of switch failures is desired.

Question 2

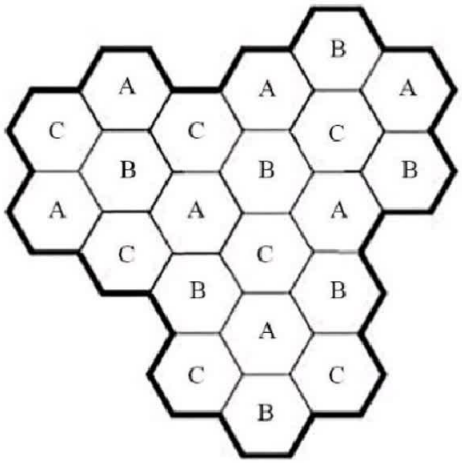
There are two cellular networks to provide call service to subscribers, the first network has hexagonal cells and the second has square cells. There are 2100 separate frequency bands and each frequency band can transmit one T1 carrier. By considering that the use of similar frequency bands is prohibited in adjacent cells,

a) what is the maximum number of frequency bands that can be assigned to each cell in the first and second networks?

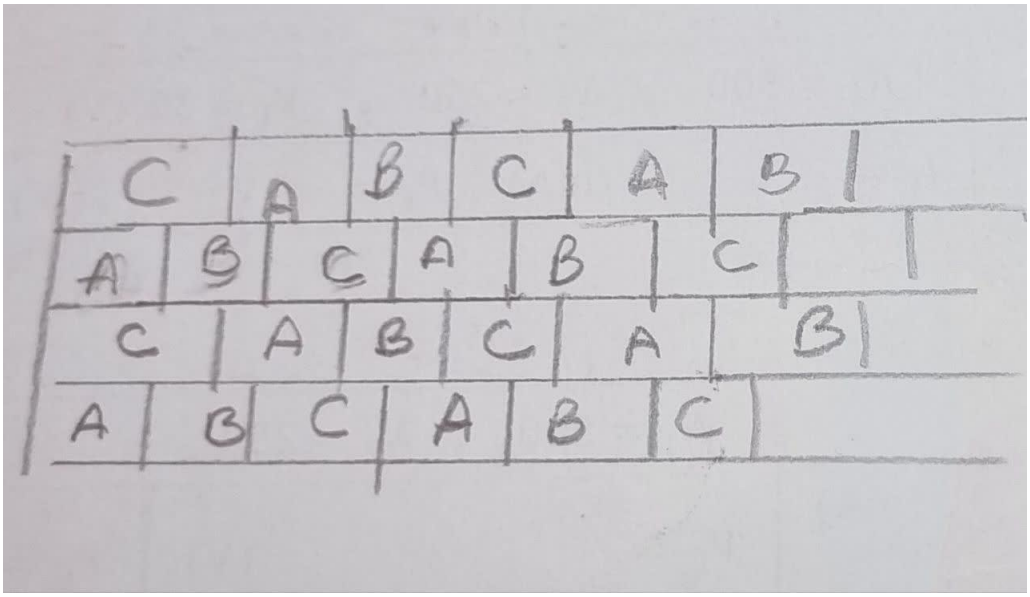
b) how many subscribers are served by one cell in the first and second network?

Solution

Hexagonal: n=3: $\frac{2100}{3} = 700$ frequency bands are allocated to each cell



Square:n= 3: $\frac{2100}{3} = 700$ frequency bands are allocated to each cell



b) Hexagonal: $700 \times 24 = 16800$ user

Square: $700 \times 24 = 16800$ user

Question 3

In figure 1, the transmitted power is 125mW, and the minimum acceptable SNR at the receiver is 30dB



Figure 1: Server and client

$$N_0 = 4 \times 10^{-21} \frac{W}{Hz}, BW = 400kHz$$

a) Suppose the server is connected to the client with a Copper cable, and the transmission loss of the line is 2 dB/km . Is it possible to have a successful connection without using any amplifier?

b) Assume that we have m identical amplifiers, each with a gain of 10dB. If your answer to the last part is no, find the minimum of m to have a successful connection.

c) Repeat the previous parts assuming the server and the client are connected with an optical fiber, and the transmission loss of the line is 0.3 dB/km

a)

$P_n = N_0 \times BW = 1.6 \times 10^{-12}\ mW$

$P_n = 10\log(1.6 \times 10^{-12}\ mW) = -117.95\ dBm$

$P_s = 10\log(125\ mW) = 20.96\ dBm$

$SNR = P_s - L - P_n = 20.96 - 200 \times 2 + 117.95 = -261.09 < 30$

It is not possible to have a successful connection without using any amplifier.

b)

$P_s - L - P_n + 10m > 30: \ -261.09 + 10m > 30$

$m > 29.1 \ \ , m_{min} = 30$

c) $SNR = P_s - L - P_n = 20.96 - 60 + 117.9 = 78.91 > 30$

It is possible to have a successful connection without using any amplifier.

Question 4

Answer the questions below about Figure 2

a) How long will it take to transmit a 1-GB file from one VSAT to another using a hub based on figure 2? Assume that the uplink is 1 Mbps, the downlink is 7 Mbps, and circuit switching is used with 1.2 sec circuit setup time.

b) Calculate the transmit time in the previous problem if packet switching is used instead. Assume that the packet size is 64 KB, the switching delay in the satellite and hub is 10 microseconds, and the packet header size is 32 bytes.

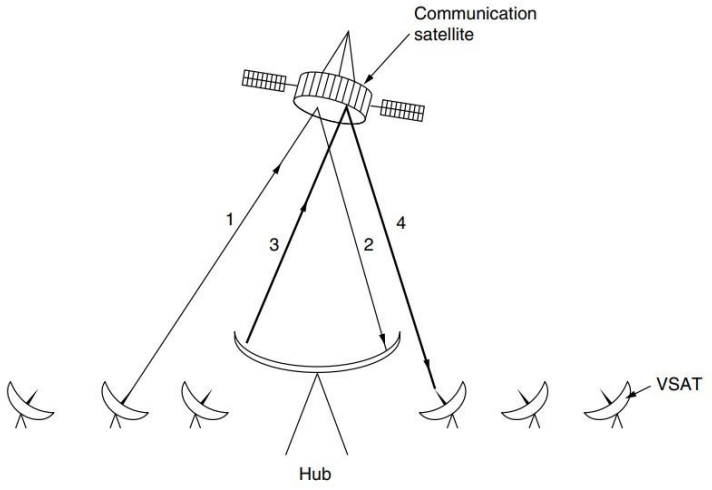


Figure 2: VSATs using a hub

Solution

1.

$$1GB = 8Gb = 8 * 10^9 bits$$

The bottleneck is the 1Mbps so the transmission time will be:

$$8 * 1024 = 8192\ s$$

The circuit switching is used so we have a circuit setup of 1.2 s. Also propagation delay is 480 msec and the whole transmission time would be:

$$8192 + 1.2 + 0.48 = 8193.68\ s$$

2. the packet size is 64kB so we have:

$$64 * 1024 = 65536$$

1Gb equals 1073741824 bits so we have:

$$packets = \frac{1073741824}{65536} = 16384$$

We also have a 32 byte header for each packet so each packet is 524544 bits now we can calculate the total transmission time. This time is calculated as below:

$$the\ bottleneck:\ (n-1) * \frac{L_{packet}}{R_{min}}$$

$$each\ packet\ transmission\ time = 2 * \left(\frac{L_{packet}}{R_{up}} + \frac{L_{packet}}{R_{Down}}\right)$$

$$propagation\ delay$$

So the final answer is:

$$total\ time = 16383 * \frac{524544}{1Mbps} + 2 * \left(\frac{524544}{1Mbps} + \frac{524544}{7Mbps}\right) + 30\ \mu s = 8196.64$$

Question 5

Suppose that users A, B, C, and D are simultaneously transmitting bits using a CDMA system, with their assigned sequences as follows:

A= (-1, -1, -1, +1, +1, -1, +1, +1)

B= (-1, -1, +1, -1, +1, +1, +1, -1)

C= (-1, +1, -1, +1, +1, +1, -1, -1)

D= (-1, X, -1, -1, Y, Z, +1, -1)

a) Find X,Y and Z.

b) Is it possible that after all 4 users have sent their bits, the output becomes (0, 0, -2, 0, -2, -4, 1 2), if the answer is yes, determine the transmitted bit of each user.

c) Is it possible that after 3 users have sent their bits, the output becomes (1, -3, 3, -1, 1, 1, 1, 1), if the answer is yes, determine the transmitted bit of each user.

Solution

a)

$D.A = 0: +1 - X + 1 - 1 + Y - Z + 1 - 1 = 0$

$D.B = 0: +1 - X - 1 + 1 + Y + Z + 1 + 1 = 0$

$D.C = 0: +1 + X + 1 - 1 + Y + Z - 1 + 1 = 0$

$X = 1$

$Y = -1$

$Z = -1$

b) $x_6 = -4: A - B - C + D = (0, 0, -2, 0, -2, -4, 2, 2)$

so it is not possible to receive (0, 0, -2, 0, -2, -4, 1 2)

c)

(1, -3, 3, -1, 1, 1, 1, 1). A=0

(1, -3, 3, -1, 1, 1, 1, 1).B / 8=1

(1, -3, 3, -1, 1, 1, 1, 1).C / 8=-1

(1, -3, 3, -1, 1, 1, 1, 1).D / 8=-1

User B sends 1 and user C and D send 0.

Question 6

A simple telephone system consists of two end offices and a single toll office to which each end office is connected by a 1-MHz full-duplex trunk. The average telephone is used to make four calls per 8-hour workday. The mean call duration is 6 min. Ten percent of the calls are long distance (i.e., pass through the toll office). What is the maximum number of telephones an end office can support? (Assume 4 kHz per circuit.) Explain why a telephone company may decide to support a lesser number of telephones than this maximum number at the end office.

Solution

Each telephone makes 0.5 calls/hour at 6 minutes each. Thus, a telephone occupies a circuit for 3 minutes/hour. 20 telephones can share a circuit. Since 10% of the calls are long distance, it takes 200 telephones to occupy a long-distance circuit full time. The interoffice trunk has 1,000,000/4000 = 250 circuits multiplexed onto it. With 200 telephones per circuit, an end office can support 200*250 = 50,000 telephones.

Supporting such a large number of telephones may result in significantly long wait times. For example, if 5,000 (10% of 50,000) users decide to make a long-distance telephone call at the same time, the wait time will be increased. This will clearly result in unhappy customers

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Question 7

Consider a simple wireless network with three pair of transmitter and receivers, shown in figure 4. T_a, T_b, T_c are transmitters and their receivers are . R_a, R_b, R_c, respectively. Suppose that A, B and C simultaneously send their data to their receivers. Also, consider that when a transmitter sends a signal with a power of P_t, the receiver gets a signal with a power of $P_r = \alpha \frac{P_t}{d^n}$ that d is the distance between the transmitter and the receiver, α is a constant value and n = 2. If P_t for $T_a, T_a \; and \; T_a$ are 1 , 2 and 3 $watt$, by considering that the noise is negligible, find the maximum data rate at R_A, R_B and R_C. ($BW = 50\ MHz$)

- Note that in the presence of interference, to obtain the maximum data rate, SINR is used instead of SNR.

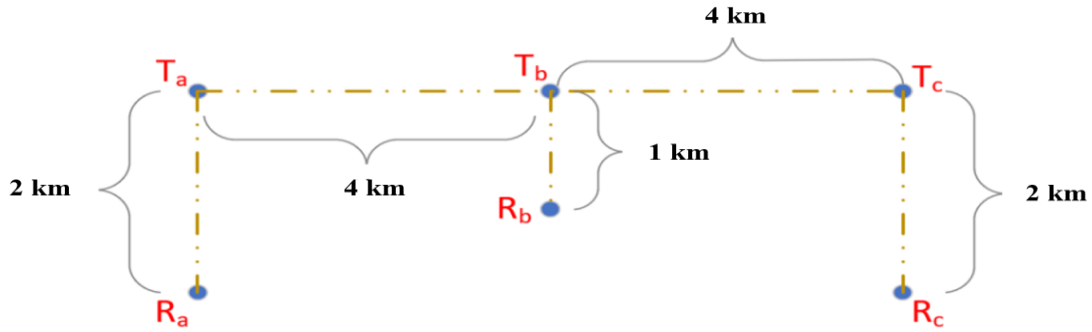


Figure 3: A network with 3 transmitters and 3 receivers

Solution

Receiver a: desired signal:

$$P_r = \alpha \frac{P_t}{d^n} = \alpha \frac{1}{2^2}$$

Interference from b and c:

$$\alpha \frac{2}{2^2 + 4^2} + \alpha \frac{3}{68}$$

Receiver b: desired signal:

$$P_r = \alpha \frac{P_t}{d^n} = \alpha \frac{2}{1^2}$$

Interference a and c:

$$\alpha \frac{1}{1^2 + 4^2} + \alpha \frac{3}{1^2 + 4^2}$$

Receiver c: desired signal:

$$P_r = \alpha \frac{P_t}{d^n} = \alpha \frac{3}{2^2}$$

Interference b and a:

$$\alpha \frac{2}{2^2 + 4^2} + \alpha \frac{1}{68}$$

Receiver a:

$rate_a = 50 \times 10^6 \times \log 2(1 + 1.73) = 72.4\ MHz$

Receiver b:

$rate_b = 50 \times 10^6 \times \log 2(1 + 8.5) = 162.4\ MHz$

Receiver c:

$rate_c = 50 \times 10^6 \times \log 2(1 + 6.53) = 145.6\ MHz$

Question 8

The figure below shows a router with a link that are part of a bigger network. Every time a packet is received at the router it is time-stamped (TA). It is also time stamped when it is about to leave link L (TB). When 5 packets of the size 200 bytes are transmitted through the link L, (TB - TA) is measured to be 6 ms, 5 ms, 2.4 ms, 4 ms, and 3.6 ms. When 5 packets of the size 500 bytes are transmitted through the link L, (TB - TA) is measured to be 11.0 ms, 6 ms, 4.2 ms, 3.8 ms, and 5.5 ms. Assume that processing delay at the router is negligible.

a) What is the average queuing delay experienced by the 200 and 500 byte packets?

b) What are the reasonable estimates of transmission and propagation delays experienced by a packet of size 1000 bytes sent through the link L?

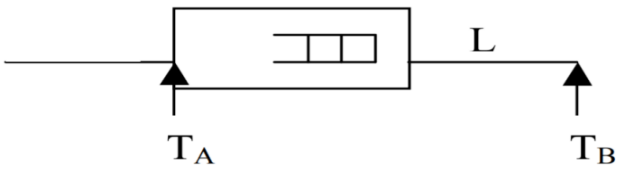


Figure 1: A router with link L

Solution

Find the minimum $T_B - T_A$:

For 200 byte packets: $\min T_B - T_A$: $2.4\ ms$

For 500 byte packets: $\min T_B - T_A$: 3.8 ms

Average queuing delay experienced by the 200 byte:

$$\frac{(6 - 2.4) + (5 - 2.4) + (2.4 - 2.4) + (4 - 2.4) + (3.6 - 2.4)}{5} = 1.8\ ms$$

Average queuing delay experienced by the 500 byte:

$$\frac{(11 - 3.8) + (6 - 3.8) + (4.2 - 3.8) + (3.8 - 3.8) + (5.5 - 3.8)}{5} = 2.3\ m$$

$trans\ rate: x\ \frac{bit}{sec}$

$prop\ delay: y$

$$\frac{200 * 8}{x} + y = 2.4\ ms$$

$$\frac{500 * 8}{x} + y = 3.8\ ms$$

$x = 1.71\ mbps\ , y = 1.5\ ms$

$$t_{tot} = \frac{1000 * 8}{x} + y = 6.1\ s$$